Copernicus Global Land Operations "Cryosphere and Water"

"CGLOPS-2"

Framework Service Contract N° 941119 (JRC)

QUALITY ASSESSMENT REPORT

LAKE ICE EXTENT (LIE)

COLLECTION 250M CONTINENTAL EUROPE

VERSION 2.2.1

Issue I1.03

Organization name of lead contractor for this deliverable: Syke

Book Captain:	Kirsikka Heinilä (Syke)
Contributing Authors:	Sari Metsämäki (Syke) Sakari Väkevä (Syke)

Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

Document Release Sheet

Book captain:	Kirsikka Heinilä	Sign	Date
Approval:	Roselyne Lacaze	Sign	Date
Endorsement:	Nadine Gobron	Sign	Date
Distribution:	Public		

Change Record

Issue/Rev	Date	Page(s)	Description of Change	Release
	03.03.2017	All	First issue	I1.01
	19.05.2017	All	Update after external review	I1.02
	19.05.2024	All	Minor updates after transferring data input sensor from MODIS to VIIRS.	I1.03

TABLE OF CONTENTS

1 Background of the document 9

1.1 Executive Summary 9

1.2 Scope and Objectives 9

1.3 Content of the document 9

1.4 Related documents 10

1.4.1 Applicable documents 10

1.4.2 Input 10

1.4.3 Output 10

2 Review of Users Requirements 11

3 Quality Assessment Method 14

3.1 Overvall procedure 14

3.2 In situ Reference Products 14

4 Results 16

5 Conclusions 19

6 Limitations and recommendations 20

7 References 21

© C-GLOPS Lot1 consortium

List of Figures

Figure 1. Map of *in situ* sites for validation of ice-breakup 16

Figure 2. Histogram of difference in days between *in situ* and satellite ice disappearance date. ... 17

Figure 3. Comparison between the difference in days and number of days from *in situ* ice disappearance observation to the date when satellite interpretation could be made. 18

© C-GLOPS Lot1 consortium

List of Tables

Table 1. Validation datasets from operational observations by Syke 15

Table 2. Percentages of open water pixels (from all visible observation area pixels) from the first satellite interpretation after the *in situ* observation 18

Table 3. Statistics of the difference between *in situ* and satellite derived ice disappearance date. The unit used is days. 19

List of Acronyms

ATDB	Algorithm Theoretical Basis Document
C-GLOPS2	Copernicus Global Land Operations Lot 2
ESA	European Space Agency
FCDR	Fundamental Climate Data Records
LIE	Lake Ice Extent
MODIS	Moderate Resolution Imaging Spectroradiometer
SCE	Snow Cover Extent
SPOT	Satellite Pour l'Observation de la Terre, "Satellite for observation of Earth"
Syke	Finnish Environment Institute
VIIRS	Visible Infrared Imaging Radiometer Suite

1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

Copernicus Global Land Operations Lot 2 (C-GLOPS2) is an operation of the Global Land Component, thematic domain Cryosphere and Water. It is allocated as a component of the Land service to ensure the ramp-up and the operational production of "Cryosphere and Water" related bio-geophysical parameters. Lake Ice Extent 250m is one of the Copernicus Global Land Service products. CEURO LIE 250m is a legacy product of LIE BALTIC 250m.

The CEURO LIE 250m classifies, on pixel basis, inland/freshwater bodies as (1) Fully snow-covered ice; (2) Partially snow-covered ice or snow-free ice; (3) Open water. A certain class is associated to a pixel following the rule of highest probability. Classification is provided only for cloud free pixels, with dedicated cloud masks for the Lake Ice Extent product. The gridded data product covers Continental Europe [72°N / 11°W to 35°N / 50°E] in WGS-84 geographical coordinate system with a resolution of 0.0025 degree (~250m, depending on latitude).

The document presents the results of the quality assessment of LIE-250m product.

1.2 SCOPE AND OBJECTIVES

In order to apply this satellite derived data product the user needs information on the uncertainty related to the interpretation and understanding how the uncertainty information was derived. The purpose of this document is to provide information for the user to evaluate the suitability of the product (in the view of performance characteristics) to the purpose in hand.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the users requirements, and the expected performance
- Chapter 3 describes the methodology for quality assessment, the metrics and the criteria of evaluation
- Chapter 4 presents the results of the analysis
- Chapter 5 summarizes the main conclusions of the study
- Chapter 6 makes recommendations based upon the results

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: GCOS – 245 : The 2022 GCOS ECVs Requirements (GCOS 245), WMO, 2022

AD4: Copernicus Global Land Component Lots 1, 2 & 3 Operation of the Global Land Component (Framework Service Contract N° 199494, 199496, 199497, 941119): Service Specifications (CGLOPS-SSD). Issue 1.00. 24th November 2016.

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service.
CGLOPS2_SVP	Service Validation Plan of the Global Land Service
CGLOPS2_PUM_LIE_250m_V2.2.1_I1.03	Product User Manual summarizing all information about the LIE-250m product

1.4.3 Output

Document ID	Descriptor
CGLOPS2_QAR_LIE_250m_V2.2.1_I1.03	Report describing the results of the scientific quality assessment of the LIE 250m product

2 REVIEW OF USERS REQUIREMENTS

According to the applicable document [AD2], the user's requirements relevant for European LIE 250m are:

- **Definition**: Continental Europe Lake Ice Extent (CEURO LIE)

- **Geometric properties**:
  - The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view.
  - Pixel coordinates shall be given for centre of pixel.

- **Geographical coverage**:
  - Projection: geographical latitude, longitude
  - Geodetical datum: WGS84, EPSG:4326
  - Pixel size: 0.0025°
  - Coordinate position: pixel centre
    - Window coordinates:
      - Upper left: 11ºW, 72ºN
      - Lower right: 50ºE, 35ºN

- **Accuracy requirements**:

  Target requirements for lake ice cover based on Global Climate Observing System (GCOS) (AD4) are following:

  - Spatial resolution: 300 m
  - Temporal resolution: daily
  - Measurement uncertainty: ≤10%

  User requirements related to LIE products are described by the GCOS – 245 : The 2022 GCOS ECVs Requirements (GCOS 245) (AD3) and in User Requirements of CryoLand Snow and Land Ice services - Cryoland (Malnes et al. 2015).

  The main user groups interested in these satellite products are the climate and hydrological modelling communities and weather forecasting authorities. The following paragraphs will summarize the thematic, temporal, and geometric aspects concerning the LIE 250m product.

Thematic aspects

The LIE 250m product classifies a section of the freshwater body as (1) Fully snow-covered ice, (2) Partially snow-covered ice/clear ice, and (3) Open water. Additionally, a class is given for fully snow-covered ice. The targeted minimum measurement uncertainty is 10% according to AD3.

Geometric aspects

Mapping the ice on lakes over the Continental Europe with a daily coverage requires at the moment the use of moderate resolution imagery due to the lack of daily high-resolution data. However, mapping of ice on small lakes requires a fine spatial resolution. The current spatial resolution for the parameter LIE is 250 m for optical data, with a geolocation accuracy of 1 pixel (Fernández-Prieto et al. 2012). NOAA-20 (JPSS-1) VIIRS (Visible Infrared Imaging Radiometer Suite) data provide good daily coverage over Continental Europe, with multiple overlapping overpasses per day.

Temporal aspects

The LIE 250 m is a daily product. Spatially and temporally consistent data sets are important for climate modelling and numerical weather prediction. Climate modellers need data sets over large areas (continental scale) and for long time periods (decades). Aggregated (gridded) data sets in space are preferred. The data sets do not need to be very frequent in time (monthly information is sufficient according to Climate modellers). For other than climatic use, Malnes et al. (2015) summarize that the majority of users request one day temporal resolution. When bridging the often wishful thinking from users with what is actually possible to achieve with current and near future satellite sensors, it should be realistic to obtain a temporal resolution of one week, due to frequent cloud coverage at high latitudes where lake ice occur. There is hence a need to also provide a level of refinement to the ice products, where temporal interpolations provide the data regularity desired.

Near real-time data:

The climate modellers have little need for real-time data: annual updates (mostly concerning ice break-up) are sufficient.

Weather forecasters may in principle need near real time data, or at least data for the latest date. The most urgent need for real time data seems to arise from within the hydrological community, for whom the real time data is a pre-requisite for flood management.

Other aspects

Satellite data on freezing and break-up dates of lake ice is also of interest to all approached users, and in particular for validation of models. These data types can also be of interest for environmental assessment and for biological monitoring.

3 QUALITY ASSESSMENT METHOD

3.1 OVERVALL PROCEDURE

The validation of LIE 250m was conducted in SEN3APP (EU FP7 No 607052) 2014–2016 project, aiming to build satellite data product services for dedicated customers. One of the products in the SEN3APP portfolio was Lake Ice Extent from optical satellite data, developed at the Finnish Environment Institute (Syke). The product performance was evaluated by comparing the date of disappearance of ice cover from the field of view from the observation point to the satellite date of ice disappearance derived from optical satellite data, in this case Terra/MODIS- satellite data. The *in situ* variable for ice disappearance is determined as the date when ice has disappeared from the observation point (field of view) entirely. The validation was carried out for 34 Finnish lakes, stretching across the country in latitude and spanning four years, namely 2010–2013. The product was validated with following steps:

1) Determine *in situ* site locations used for validation
2) Extract a time-series of sub-regions of LIE- data product from radius of 2.0 km around the *in situ* observation site
3) Extract the days when the observation areas are not totally covered by clouds (this omits the possibility of ice underneath the clouds).
4) Compare the *in situ* observation date and the satellite observation date (calculate statistical measures; average, STD and median)

Results from this validation exercise are given in the following sections.

3.2 IN SITU REFERENCE PRODUCTS

Syke has unique *in situ* datasets of lake ice observations dating back to the end of 19th century. The observations are still operational. The ice-off observations are performed by Syke on daily basis during the ice season (VHJ 1984). Temporally the dataset covers the satellite era. The spatial domain is Finland with distribution covering the whole country. Table 1 below lists the datasets available for validation. The LIE product has high uncertainty during the lake freezing period due to low light conditions in the higher latitudes. Currently the product is only validated against ice break-up observations.

Table 1. Validation datasets from operational observations by Syke

In situ dataset	*Freezing and breakup*	*Ice thickness and snow on ice thickness*
Unit	Date	cm
Accuracy	±1 day	±1 cm
Observation interval	Daily observations	3 per month (10th, 20th and 30th)
Areal extent	Entire observation period: 122 sites	Entire observation period: 161 sites
Temporal coverage	1753–2015 (changing network)	1912–2015 (changing network)

4 RESULTS

Validation against in situ observations of ice break-up

Currently only ice-breakup is used for validation as there is still large uncertainty in the freezing period of lakes due to low light conditions in the northern latitudes in autumn due to polar night. Validation of LIE product to estimate ice break-up is based on four years of LIE observations and *in situ* observations conducted operationally by Syke. The four-year dataset shows good performance of the methodology considering its simplicity. The validation sites are shown in Figure 1. Although the validation is only conducted in Finnish lakes, the sites spread across wide range of latitudes and different types of lakes.

The histogram in Figure 2. shows that in year 2011 the difference to the other years was fairly significant. This is due to long period of cloud cover in late April – early May, where most *in situ* observations of ice disappearance occur. This can be seen in the scatterplot in Figure 3. where the difference in days is plotted against the number of days after the *in situ* ice disappearance observation before the next satellite observation.

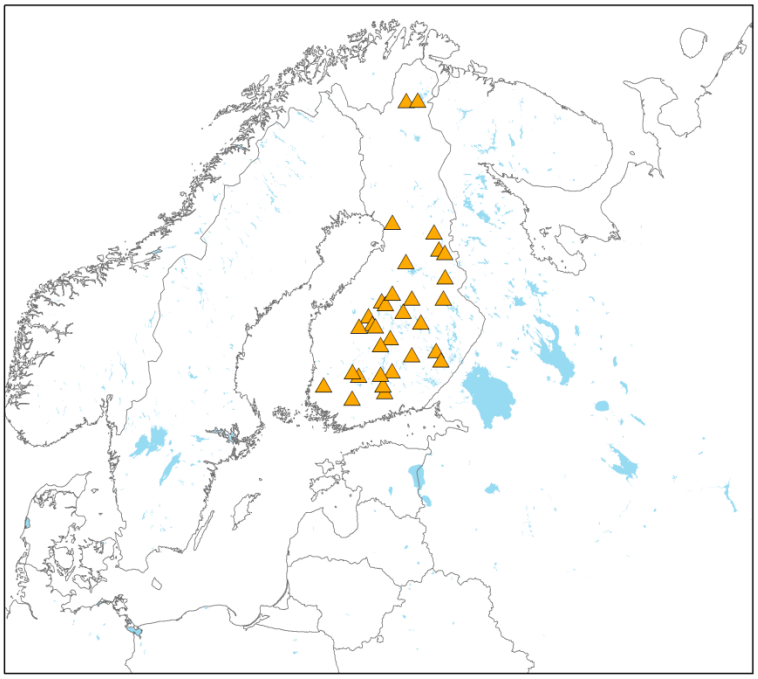


Figure 1. Map of *in situ* sites for validation of ice-breakup

The satellite method has tendency to produce later ice disappearance date than *in situ* observation. This should be the case as in many cases there is some gap between the *in situ* observation day

and the next day when satellite data is available from the observation area. The analysis here does not take into account the fact that the water mask covers only part of the field of view from the observation point, meaning, that there can be ice close to the shore, but out of the interpretation area of the satellite, producing too early ice clearance for the site from satellite data. Here the interpretation also takes into account only the cloud-free area. Therefore there is possibility of ice underneath the cloud cover, again producing too early day for ice disappearance. Undetected cloud cover, changing water properties (in heavily turbid waters) and changes in reflectance due to sun glint effects are the main causes of overestimation in the ice cover, producing later disappearance of ice cover from satellite data.

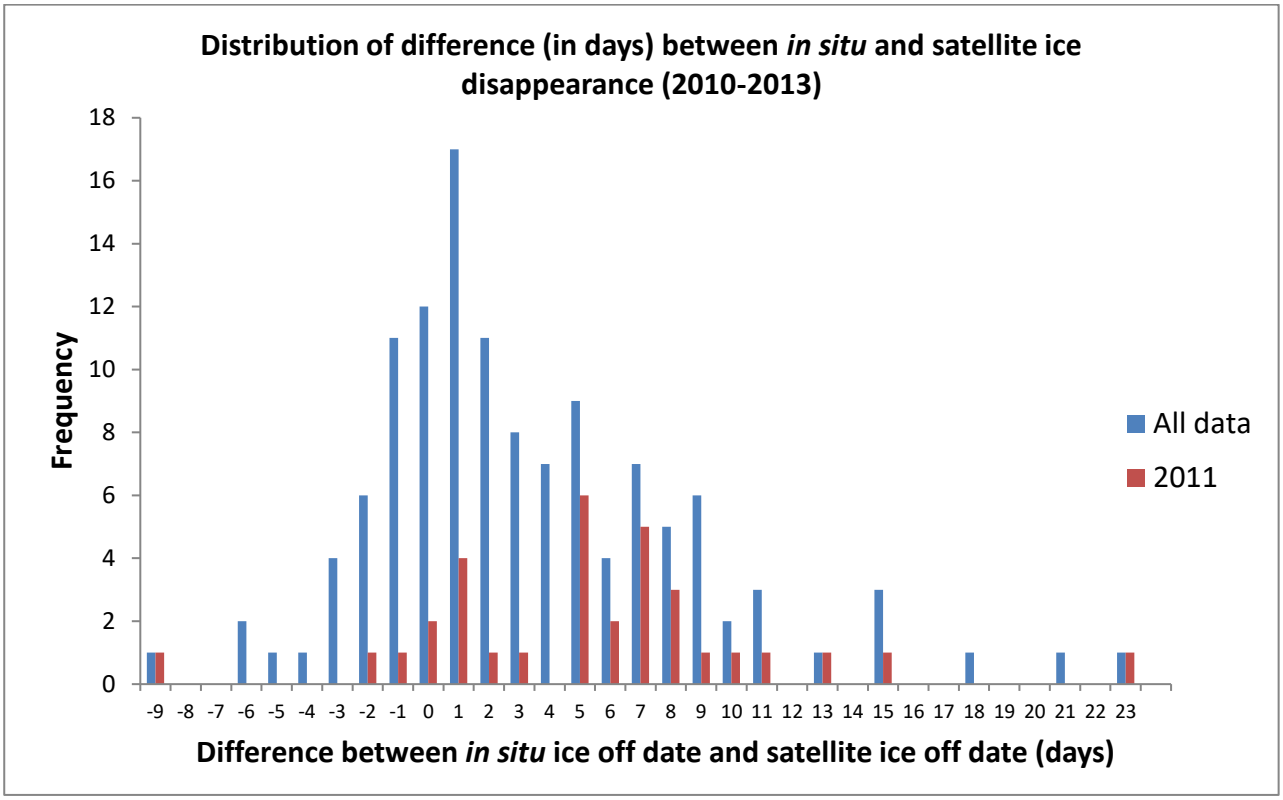


Figure 2. Histogram of difference in days between *in situ* and satellite ice disappearance date.

To further evaluate the performance of the product, especially in the case of positive difference between satellite and *in situ* derived data (i.e. satellite interpretation being later than *in situ*), the next available satellite observation from *in situ* ice disappearance was interpreted for the percentage of open water pixels from the visible pixels (not cloud-covered) for the observation site. The data shows that in over 50% of the cases during 2010-2013 the next satellite interpretation was showing that the ice had disappeared (Table 2).

Table 2. Percentages of open water pixels (from all visible observation area pixels) from the first satellite interpretation after the *in situ* observation.

Open water percentage [%]	Number of cases	Percentage of cases (N=124) [%]
0 ≤ x ≤ 33	31	25.0
33 ≤ x < 67	11	8.8
67 ≤ x <100	17	13.7
x = 100	65	53.4

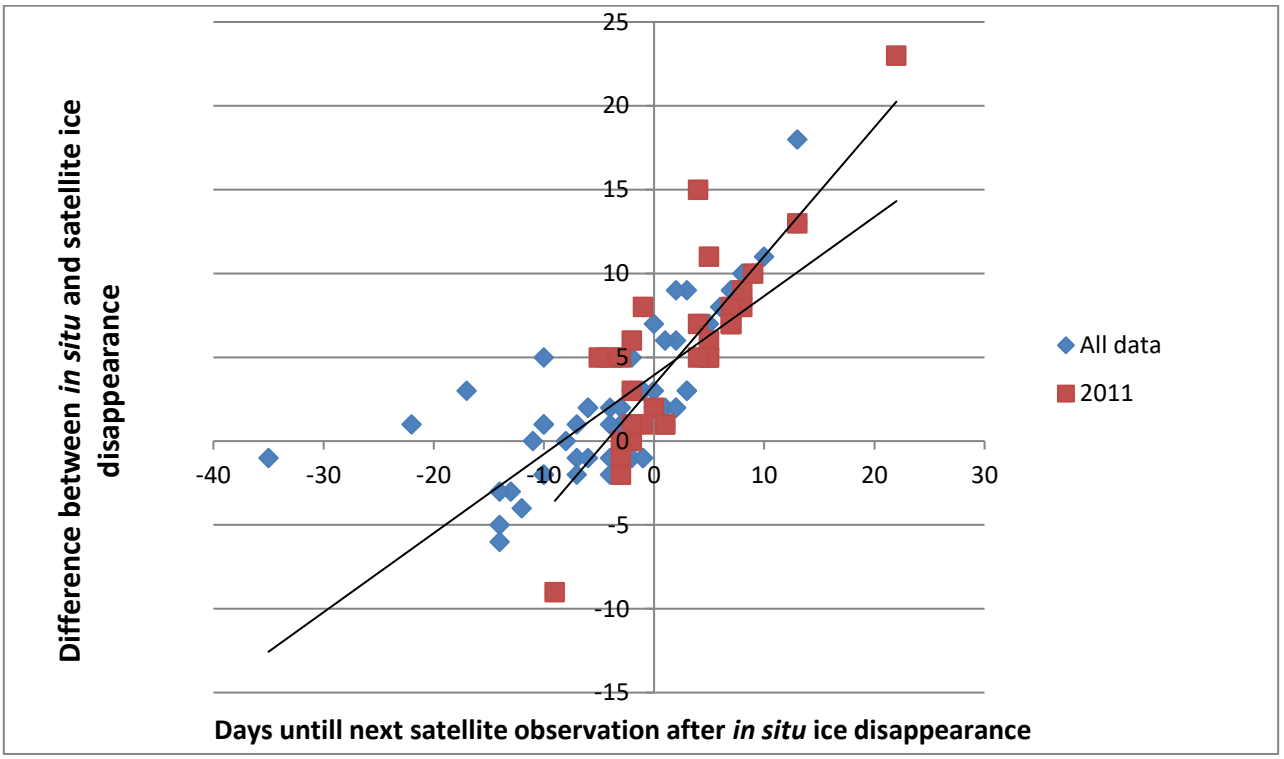


Figure 3. Comparison between the difference in days and number of days from *in situ* ice disappearance observation to the date when satellite interpretation could be made.

5 CONCLUSIONS

Here the validation has been done in the context of comparing *in situ* observation of disappearance of lake ice to the same parameter derived from satellite data. Considering the simple implementation and adaptation of the method for different optical satellite sensors, the performance of the method was good. Table 3 shows the average difference between *in situ* and satellite observations and the standard deviation of the difference for the four years, 2010-2013, used for validation. This shows that on average the difference is around +3 days and median of the dataset being +2 days, the standard deviation being 5.2 days. In future, the Lake Ice Extent 250m product will be spatially extensively validated against high resolution LIE products derived from optical satellite data of Sentinel-2, with Landsat-8 as back-up applying procedures and protocols developed in the EU FP7 project CryoLand, including statistical information.

Table 3. Statistics of the difference between *in situ* and satellite derived ice disappearance date. The unit used is days.

	2010	2011	2012	2013	All
Average	3.4	5.4	2.5	1.8	**3.3**
Median	2	5	2	1	**2**
STD	5.8	5.5	4.4	4.1	**5.2**

6 LIMITATIONS AND RECOMMENDATIONS

Continental Europe Lake Ice Extent (CEURO LIE) is monitored daily using optical satellite data. Observations from optical satellite sensors are restricted by cloud cover as well as low light conditions from late autumn to early spring in the high northern latitudes due to the polar night.

Mapping of ice on lakes requires, especially in Fennoscandia, fine spatial resolution watermask due to the small size of the lakes and relatively high number of islands and narrow parts in some lakes. The gridded CEURO LIE data product covers the Continental Europe (see geographical coverage above) with 0.0025° (~250m) resolution. The shores (also in case of island) are not included in the product to avoid mixed pixels, i.e., containing both land and water/ice). The user should be aware that during the freezing and break-up of ice the method cannot always detect whether the target represents water on dark (thin, semitransparent) ice or open water. Same goes for fact that the small white ice floes within a single pixel may results the pixel being classified as ice. However, in these cases the lake ice is already fragmented and/or dark and weak and thus does not always cause disadvantage for the end users depending on the user case.

Validation against high resolutions satellite data from Sentinel-2 and Landsat-8 are foreseen, to provide further information about the performance of the product.

7 REFERENCES

Fernández-Prieto, D., Duguay, C., Gauthier, Y., Gustafsson, D., Malnes, E., Mattila, O.-P., Rontu, L., Rott, H., Samuelsson, P., & Solberg, R. (2012). ESA STSE North Hydrology: Development of multi-mission satellite data products in support of atmospheric and hydrological modeling of cold regions. In, *EGU General Assembly*. Vienna, Austria

IGOS: Cryosphere Theme Report 2007, WMO/TD-No 1405,World Meteorological Organization, Geneva, Switzerland, 100 pp.,2007.

Malnes, E., Buanes, A., Nagler, T., Bippus, G., Gustafsson, D., Schiller, C., Metsämäki, S., Pulliainen, J., Luojus, K., Larsen, H.E., Solberg, R., Diamandi, A., & Wiesmann, A. (2015). User requirements for the snow and land ice services – CryoLand. *The Cryosphere, 9*, 1191-1202

VHJ (1984). Hydrologiset havainto- ja mittausmenetelmät. *Publications of the National Board of Waters, Finland*